

AI Basics Starter Guide

A fast, student-friendly map of the ideas that show up most often in AI club conversations.

What this guide is for

- Use this when someone is brand new and needs the big picture first.
- The goal is conceptual clarity, not deep math.
- If a term here sounds vague, treat it as a starting definition and then follow the source trail.

1. The big stack

The easiest way to keep AI terms straight is to see them as a stack, not as random buzzwords.

Layer	Plain-language meaning
Artificial intelligence (AI)	The broad field: making systems do tasks that usually require human-like perception, prediction, generation, or decision support.
Machine learning (ML)	A major part of AI. Instead of hand-writing every rule, you train a model to learn patterns from data.
Deep learning	A major part of ML. It uses neural networks with many layers and is behind most modern image, speech, and language systems.
Generative AI	Systems that produce new content - such as text, images, audio, video, or code - from patterns learned during training.
Large language model (LLM)	A generative AI model for language. Modern LLMs work with tokens, use context, and rely on transformer-style attention.

2. How a model actually gets used

At a high level, modern AI systems follow a simple loop:

1. Collect or choose data that represents the task well.
2. Train a model so it adjusts its internal weights to do better on examples.
3. Evaluate the model on new cases instead of only the training cases.
4. Deploy the model for inference - meaning it now answers or predicts on fresh input.
5. Monitor quality, failures, bias, and misuse, then improve the system.

Two words students mix up a lot

- Training = the learning phase, where the model changes its parameters.
- Inference = the use phase, where the trained model responds to a new input.

3. Key ideas that matter most

Term	What it means
Data	Examples the model learns from. Bad data usually leads to bad model behavior.
Feature	A measurable property used by a model. In classic ML, people often design features by hand.
Neural network	A model built from layers of units that transform inputs into outputs.
Token	A chunk of text a language model works with - often a word piece, not always a full word.
Prompt	The input instructions or context given to a generative model.
Context	The surrounding information the model can use when producing the next output.
Embedding	A numerical representation that places related items closer together in vector space.
Fine-tuning	Additional training to adapt a model to a narrower task or style.
Hallucination	A fluent-sounding output that is false, unsupported, or invented.
Benchmark	A standard test used to compare systems on a defined task.
Bias	A systematic pattern of unfairness, distortion, or skew in data, modeling, or outputs.

4. What makes LLMs feel different

- They work on tokens, not whole ideas all at once.
- They use context heavily: nearby words and earlier conversation matter.
- Transformer attention helps the model weigh which parts of the context matter most.
- They are trained to predict likely next tokens, which is why they can write coherent text.
- That same mechanism can also produce confident errors if the model lacks grounding or good checks.

5. Fast concept map

Mental map

- AI is the umbrella.
- ML is one major way to do AI.
- Deep learning is one major way to do ML.
- Generative AI is a family of systems that creates new content.
- LLMs are one important kind of generative AI focused on language.
- Prompts shape the input; tokens are the working units; inference is the output phase.

6. Good student habits

- Do not treat smooth wording as proof of correctness.
- Separate 'what the model said' from 'what the source says.'
- When a claim matters, ask for evidence, sources, or a way to verify it.
- If a definition sounds fuzzy, rewrite it in your own words and test it with one concrete example.
- Try to learn concepts as contrasts: training vs inference, token vs word, AI vs ML, model quality vs model confidence.

Sources used

- Google for Developers - Machine Learning Glossary (updated 2025)
- Google for Developers - Machine Learning Crash Course: Large Language Models / What's a large language model? (updated 2025)
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- IBM Think - What is Machine Learning?
- IBM Think - What is Generative AI?
- NIST - AI Risk Management Framework
- NIST - AI RMF: Generative AI Profile (2024)